Unit 6 Pre-Test

Quadratic Expressions & Equations

Standards assessed: NC.M1.A-APR.1 · A-APR.3 · A-SSE.3 · A-REI.4 · F-IF.8a

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.A-APR.1 · NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 Multiply and add polynomials, factor out a GCF, and solve simple quadratics.

• *I can multiply and add polynomials, factor out a common factor, and solve with square roots or zero products.*

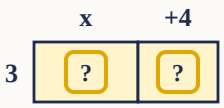
Answer each question. Use the picture to help you.

How to type your answers

power `x^2+5x+6` factored `(x+3)(x+4)` two answers `3, -5` ± `-2+sqrt(5)`

1. *Distribute*

Multiply: 3(x + 4)

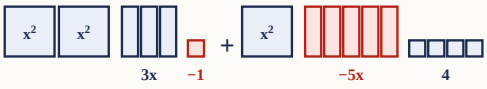


Answer: _____

answer: 3x + 12

2. *Add polynomials*

$(2x^2 + 3x - 1) + (x^2 - 5x + 4)$



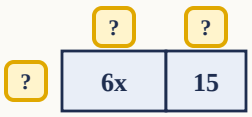
Answer: _____

answer: $3x^2 - 2x + 3$

3. *Factor (GCF)*

Factor out the greatest common factor:

6x + 15

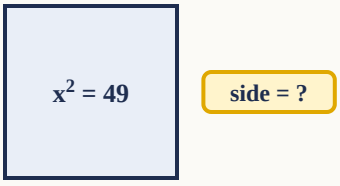


Answer: _____

answer: 3(2x + 5)

4. *Square roots*

Solve: $x^2 = 49$

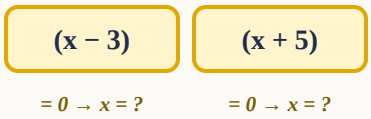


x = _____

answer: x = 7 or x = −7 (±7)

5. *Zero product*

Solve: $(x - 3)(x + 5) = 0$

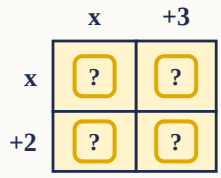


x =

answer: x = 3 or x = −5

6. *Multiply*

$(x + 3)(x + 2)$



Answer:

answer: $x^2 + 5x + 6$

SCORE 4.0 Target Content

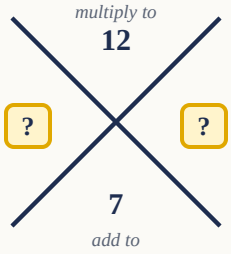
NC.M1.A-APR.1 · NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 Factor trinomials and differences of squares, and solve by factoring.

• *I can factor $x^2 + bx + c$ and $a^2 - b^2$, multiply binomials, and use factors to find zeros.*

Solve each problem. Show your work on paper.

7. *Factor*

Factor: $x^2 + 7x + 12$

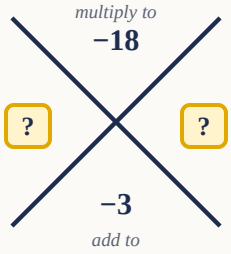


Answer:

answer: (x + 3)(x + 4)

8. *Factor*

Factor: $x^2 - 3x - 18$

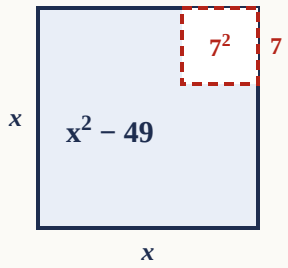


Answer:

answer: (x − 6)(x + 3)

9. *Difference of squares*

Factor: $x^2 - 49$

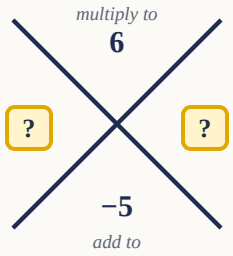


Answer:

answer: $(x + 7)(x - 7)$

10. *Solve by factoring*

Solve: $x^2 - 5x + 6 = 0$

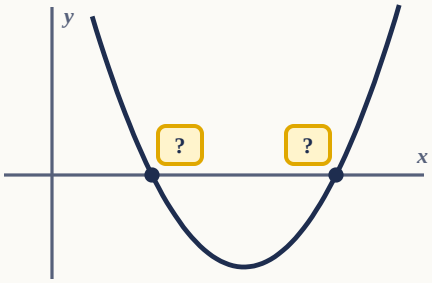


x =

answer: $x = 2$ or $x = 3$

11. *Zeros*

Find the zeros of $f(x) = (x + 1)(x - 4)$.

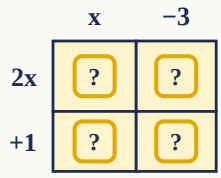


x =

answer: $x = -1$ and $x = 4$

12. *Multiply*

$(2x + 1)(x - 3)$



Answer:

answer: $2x^2 - 5x - 3$

SCORE 5.0 Advanced Content

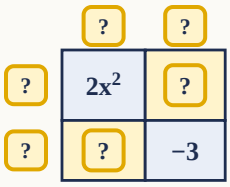
NC.M1.A-APR.3 · NC.M1.A-SSE.3 · NC.M1.A-REI.4 · NC.M1.F-IF.8a Factor with a ≠ 1, complete the square, and use the quadratic formula.

• *I can factor harder trinomials, complete the square, use the quadratic formula, and use forms to find key features.*

Solve each problem. Give every solution.

13. *Factor ($a \neq 1$)*

Factor: $2x^2 + 5x - 3$

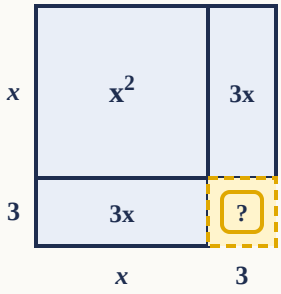


Answer:

answer: $(2x - 1)(x + 3)$

14. *Complete the square*

Solve by completing the square: $x^2 + 6x = 16$



$x =$

answer: $x = 2$ or $x = -8$

15. *Quadratic formula*

Solve: $x^2 + 4x - 1 = 0$. (Exact, or round to the hundredth.)

$$a = 1,\ b = 4,\ c = -1$$

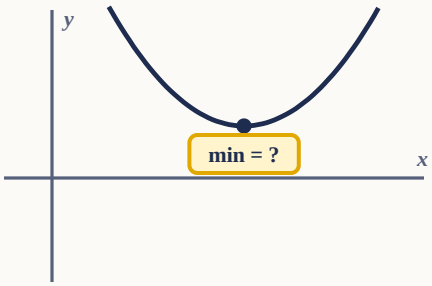
$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

$x =$

answer: $x = -2 \pm \sqrt{5}$ (≈ 0.24 and -4.24)

16. *Vertex form*

$x^2 - 6x + 11 = (x - 3)^2 + 2$. What is the **minimum value** of $y = x^2 - 6x + 11$?

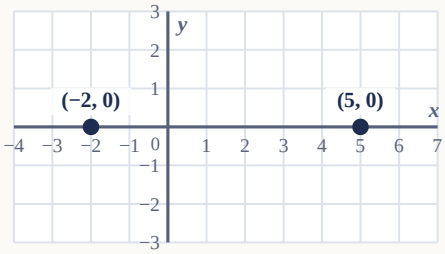


min =

answer: 2

17. *Build from zeros*

A quadratic has zeros x = −2 and x = 5, and a = 1. Write it in standard form.

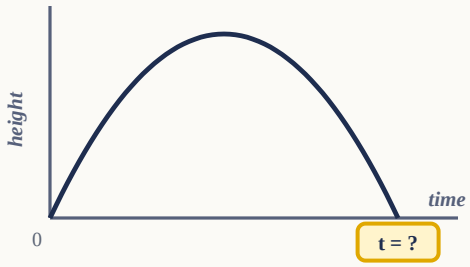


y = _____

answer: $x^2 - 3x - 10$

18. *Factored form*

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$. After how many seconds does it land ($t > 0$)?



t = _____

answer: 2 seconds